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CSE430 , Assignment 2

* Solve the following equation -

$$y(n) - 5y(n-2) + 6y(n-3) = x(n) - x(n-2)$$

The system is initially relaxed.

for impulse input, steady state response $y_s[n] = 0$

Now, to determine zero input solution -

$$\lambda^n - 5\lambda^{n-2} + 6\lambda^{n-3} = 0$$

$$\Rightarrow \lambda^n \left(1 - \frac{5}{\lambda^2} + \frac{6}{\lambda^3}\right) = 0$$

$$\Rightarrow \lambda^n \left(1 - \frac{5}{\lambda^2} + \frac{6}{\lambda^3}\right) = 0$$

$$\Rightarrow \frac{\lambda^3 - 5\lambda + 6}{\lambda^3} = 0$$

$$\Rightarrow \lambda^3 - 5\lambda + 6 = 0$$

$$\therefore \lambda = -2.7$$

$$\therefore y_n[n] = -c_1 (2.7)^n$$

$$y[n] = -c_1 \, 2.7^n \quad \text{--- (1)}$$

Given,

$$y(n) = x(n) - x(n-2) + 5y(n-2) - 6y(n-3) \quad \text{—— (ii)}$$

From → (i) =

$$y[0] = c_1$$

From (ii) →

$$y[0] = x(0) - x(-2) + 5y(-2) - 6y(-3)$$

$$= \delta(0) - \delta(-2) + 5y(-2) - 6y(-3)$$

$$= 1 \quad \text{[since initially relaxed]}$$

$$\therefore c_1(0) = 1. \quad \text{—— (iii)}$$

Again from (i) →

$$y(1) = -2.7c_1 \quad \text{—— (iv)}$$

From (ii) →

$$y(1) = x(1) - x(-1) + 5y(-1) - 6y(-2)$$

$$= \delta(1) - \delta(-1) + 5y(-1) - 6y(-2)$$

$$= 0$$

$$\therefore -2.7c_1 = 0$$

$$\Rightarrow c_1(1) = 0 \quad \text{—— (v)}$$

from equation (iii) and (v) we get -

$$C_1 = \frac{1}{2}.$$

$$\therefore \quad y(n) = \frac{1}{2}(-2.7)^n$$

(Ans)